WORKED SOLUTION

STEP I Specimen paper, Question 7

Pure • Complex Exponential Series

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s1-q7

The key idea

Whenever you see $1 + e^{i\theta}$, take out the "half-angle" factor $e^{i\theta/2}$: what is left is $e^{-i\theta/2} + e^{i\theta/2}$, which is real. For the series, the binomial coefficients $\frac{n!}{r!(n-r)!} = \binom{n}{r}$ suggest the binomial theorem. So we write each sine as the imaginary part of an exponential, and the sum becomes the imaginary part of a power of $1 + e^{2i\alpha}$, which the first part has already put into polar form.

Solution

Modulus and argument

Taking out a factor $e^{i\alpha}$,

$$\begin{aligned} 1 + e^{2i\alpha} &= e^{i\alpha}(e^{-i\alpha} + e^{i\alpha}) \\ &= e^{i\alpha}[(\cos\alpha - i\sin\alpha) + (\cos\alpha + i\sin\alpha)] \\ &= 2\cos\alpha\, e^{i\alpha}. \end{aligned}$$

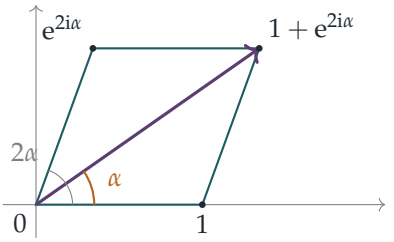
Ask yourself Can we now just read off "modulus $2\cos\alpha$, argument α"? What is the restriction on α for?

The form $r e^{i\theta}$ gives modulus r and argument θ only when $r > 0$. Here $-\frac{1}{2}\pi < \alpha < \frac{1}{2}\pi$, so $\cos\alpha > 0$, and this range also lies inside the principal interval $(-\pi, \pi]$. Hence

$$|1 + e^{2i\alpha}| = 2\cos\alpha, \qquad \arg(1 + e^{2i\alpha}) = \alpha.$$

(If $\cos\alpha$ were negative, the modulus would be $-2\cos\alpha$ and the argument would shift by π.)

There is a nice geometric picture too. The points 0, 1 and $1 + e^{2i\alpha}$ form a rhombus with sides of length 1, together with the point $e^{2i\alpha}$. The diagonal from 0 bisects the angle 2α and has length $2\cos\alpha$.



The series

Since $\sin\theta = \operatorname{Im} e^{i\theta}$ for real θ, and the binomial coefficients are real,

$$S = \sum_{r=0}^{n} \binom{n}{r} \sin(2r+1)\alpha = \operatorname{Im} \sum_{r=0}^{n} \binom{n}{r} e^{i(2r+1)\alpha}$$
$$= \operatorname{Im}\left[e^{i\alpha} \sum_{r=0}^{n} \binom{n}{r} (e^{2i\alpha})^r\right] = \operatorname{Im}\left[e^{i\alpha}(1+e^{2i\alpha})^n\right],$$

using $e^{i(2r+1)\alpha} = e^{i\alpha}(e^{2i\alpha})^r$ and then the binomial theorem. Now use the first part, with de Moivre's theorem $(e^{i\alpha})^n = e^{in\alpha}$:

$$e^{i\alpha}(1+e^{2i\alpha})^n = e^{i\alpha}\, 2^n \cos^n\alpha\, e^{in\alpha} = 2^n \cos^n\alpha\, e^{i(n+1)\alpha}.$$

Because $2^n \cos^n \alpha$ is real, taking imaginary parts gives

$$\boxed{\sum_{r=0}^{n} \frac{n!}{r!\,(n-r)!} \sin(2r+1)\alpha = 2^n \cos^n\alpha\, \sin(n+1)\alpha.}$$

The factorisation $1 + e^{2i\alpha} = 2\cos\alpha\, e^{i\alpha}$ holds for *every* real α. We needed $\cos\alpha > 0$ only to name the modulus and argument, so the sum formula is valid for all real α, not just the range given.

Checks

- $n = 0$: both sides are $\sin\alpha$. $n = 1$: the left side is $\sin\alpha + \sin 3\alpha = 2\sin 2\alpha\cos\alpha$ by the sum-to-product formula, which matches $2\cos\alpha\sin 2\alpha$.
- $n = 3$, $\alpha = \frac{\pi}{6}$: the left side is $\frac{1}{2} + 3 + \frac{3}{2} - \frac{1}{2} = \frac{9}{2}$, and the right side is $8\left(\frac{\sqrt{3}}{2}\right)^3 \cdot \frac{\sqrt{3}}{2} = \frac{9}{2}$.
- $\alpha = \frac{1}{2}\pi$: every term $\sin(2r+1)\frac{\pi}{2} = (-1)^r$, so the sum is $(1-1)^n = 0$ for $n \geq 1$, matching $\cos\frac{\pi}{2} = 0$.
- We checked numerically that the formula holds for $0 \leq n \leq 24$ at random values of α.

Common slips

- Giving the modulus as $2\cos\alpha$ without noting that the range of α makes it positive. A modulus can't be negative.
- Taking imaginary parts too early. $\operatorname{Im}(zw) \neq \operatorname{Im} z \cdot \operatorname{Im} w$, so keep everything complex until the very end.
- Writing $\mathrm{e}^{\mathrm{i}(2r+1)\alpha}$ as $\left(\mathrm{e}^{2\mathrm{i}\alpha}\right)^r$ and losing the extra factor $\mathrm{e}^{\mathrm{i}\alpha}$. That turns $\sin(n+1)\alpha$ into $\sin n\alpha$.

Beyond STEP

The same idea, summing $\sum \binom{n}{r}\mathrm{e}^{\mathrm{i}r\theta}$ in closed form, appears in Fourier analysis. There, $\cos^n$-type expressions are expanded into, or collapsed from, sums of multiple angles, and the binomial weights $\binom{n}{r}/2^n$ become the binomial distribution. Factoring out the half-angle is also the standard way to find the modulus of $1-\mathrm{e}^{\mathrm{i}\theta}$, which you meet in geometric series and roots of unity. Needham's *Visual Complex Analysis* develops the geometric view of these manipulations beautifully, rhombus picture included.

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